

A Feasibility-Aware HUBO/QUBO Benchmark for UAV Obstacle-Avoidance with Visibility Cost

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Abstract

Quantum-assisted UAV path planning has recently been studied using QAOA-style obstacle-avoidance formulations, but a low-energy binary sample is not necessarily a valid UAV trajectory. This paper presents a feasibility-aware benchmark for a discrete 8×8 , $L = 20$ UAV grid-planning problem with obstacle avoidance, a line-of-sight visibility cost, and a higher-order temporal buffer-risk term. The horizon L denotes discrete planning layers, not physical seconds. The model uses binary variables $x_{t,v}$ indicating whether the UAV occupies cell v at time layer t and forms a HUBO objective with one-hot, start, motion, obstacle, visibility, terminal-goal, and sustained near-obstacle proximity terms. The instance has 1,280 original binary variables and 118,344 polynomial terms, including 4,392 cubic monomials. We compare A*, RRT-best, a QUAU-style QAOA candidate-path selector, native trajectory-level simulated annealing (SA), the classical `neal` BQM sampler after HUBO-to-QUBO reduction, and an exact CP-SAT/MILP baseline that minimizes the native HUBO soft objective over hard-feasible paths. The CP-SAT solver returns OPTIMAL and certifies that the best observed SA path is globally optimal for the enforced feasible-path objective, with native HUBO energy $-20,752.43$, soft cost 247.57, path length 14, six turns, and 85% visibility. The results do not claim quantum speedup; they show that feasibility-aware decoding, exact baselines, and quadratization accounting are essential for interpreting quantum-compatible UAV planning results.

Introduction

UAV path planning in cluttered environments requires at least two safety properties: the path must avoid obstacles, and the decoded plan must be executable under the allowed motion model. When perception or tracking is relevant, the planner may also prefer cells with line of sight to the target. Recent quantum-assisted work such as QUAU formulates UAV obstacle-avoidance path planning using QAOA-style optimization and compares against A* and RRT (Innan et al. 2025). QUAU is therefore an important closest reference for quantum-assisted UAV navigation. However, a QAOA, annealing, or QUBO solver returns a binary sample, and the sample must be decoded and checked before it can be called a valid path.

This paper studies a deliberately compact grid abstraction so that modeling and solver behavior can be inspected directly. The goal is not to claim quantum speedup

on a small static grid. The goal is to define a reproducible HUBO/QUBO-compatible benchmark, separate energy minimization from actual path validity, and compare classical, quantum-inspired, and quantum-compatible solver pathways on the same problem instance.

Problem Statement and Scope

The planning problem considered here is a static, discrete visibility-aware path-planning abstraction. A UAV must move from a fixed start cell to a fixed target/goal cell through legal grid transitions while avoiding obstacle cells. Visibility is represented as a soft line-of-sight cost from each candidate UAV cell to the fixed target cell. Thus, this paper does not solve continuous quadrotor control, full moving-target tracking, or camera field-of-view control. The abstraction is intentionally small enough that the encoded binary model, decoded paths, feasibility failures, and exact classical baselines can be inspected in detail. The time horizon L is a design parameter giving the number of discrete planning layers. It is not assumed to be seconds unless a physical sampling time Δt is separately assigned.

The contributions are as follows. First, we give a complete binary decision-variable formulation for UAV grid planning with obstacle avoidance and a visibility cost. Second, we identify the real source of higher-order structure: a temporal buffer-risk term, not the static line-of-sight cost. Third, we introduce a feasibility-aware evaluation protocol that reports structural feasibility, goal arrival, and full-task success separately. Fourth, we implement a same-platform QUAU-style QAOA candidate-path baseline, clearly not as official QUAU code but as a reproducible reference following the published workflow. Fifth, we add exact CP-SAT and MILP implementations, enabling global comparison against the native HUBO soft objective over hard-feasible paths.

The novelty should be interpreted narrowly. This paper does not introduce a faster planner than A* or CP-SAT, and it does not demonstrate quantum advantage. Its novelty is the feasibility-aware benchmarking layer around a quantum-compatible UAV HUBO/QUBO formulation: the same decoded paths are compared under graph search, candidate-path QAOA, native trajectory-SA, quadratized BQM sampling, and exact CP-SAT/MILP baselines, with validity and goal arrival separated from energy. Without exact baselines and decoded-path checks, low-energy quantum-compatible

samples can be misleading.

Related Work

Classical and Visibility-Aware UAV Planning

Classical UAV planning spans graph search, sampling-based planning, trajectory optimization, and perception-aware control. A* remains a standard shortest-path reference on discretized maps (Hart, Nilsson, and Raphael 1968), while RRT provides a widely used sampling-based alternative (LaValle 1998). Prior work considers moving-target tracking in cluttered environments, occlusion-aware multi-robot three-dimensional tracking, aggressive target tracking with collision and occlusion avoidance, and perception-aware model-predictive control (Chen, Liu, and Shen 2016; Hausman et al. 2016; Penin, Robuffo Giordano, and Chaumette 2018; Falanga et al. 2018). Visibility-constrained tracking has also been extended to multiple UAVs (Wang et al. 2022), while real-time multi-convex MPC has been developed for occlusion-free target tracking with collision avoidance (Masnavi et al. 2022). These continuous-control studies motivate the visibility term used here, but our problem is deliberately a static discrete benchmark rather than a moving-target control formulation.

Quantum and QUBO Formulations for UAV Optimization

Quantum optimization for routing has developed rapidly. QUBO provides a standard binary optimization representation (Glover, Kochenberger, and Du 2019), while QAOA is a widely used variational gate-based approach (Farhi, Goldstone, and Gutmann 2014). A systematic review of quantum computing for routing highlights recurring issues of encoding size, constraints, and fair classical comparison (Osaba, Villar-Rodriguez, and Oregi 2022). UAV-specific studies use variational quantum algorithms to select collision-avoiding routes from predefined alternatives (Huang et al. 2022); formulate constrained multi-UAV mission planning as MILP and QUBO (Davies and Kalidindi 2024); compare QUBO drone routing with quantum, digital, and classical optimization (Hua et al. 2024); and examine feasibility and scalability in drone-delivery packing (Tarquini et al. 2026).

More recent frameworks move closer to the present setting. Q4DR combines QAOA clustering with annealing-based routing under asymmetric costs, forbidden arcs, and charging constraints (Osaba et al. 2025). QUAV uses QAOA-style optimization for single-UAV obstacle avoidance and reports simulation and quantum-hardware experiments (Innan et al. 2025). Multi-target UAV path planning has also been formulated as QUBO and solved using a coherent Ising machine (Qiu et al. 2025), while target tracking with obstacle-distance constraints has been expressed as constrained quadratic models for quantum annealing (Barbeau, Janabi-Sharifi, and Masnavi 2025). These studies demonstrate multiple quantum-compatible routes for UAV routing, collision avoidance, and tracking, but they optimize different abstractions and generally do not center evaluation on decoded structural feasibility under a common native objective.

QUBO Path Finding and the Benchmarking Gap

Recent multi-agent path-finding studies are relevant to encoding and scalability. Hybrid quantum-classical MAPF can use branch-and-cut-and-prize with iteratively solved QUBO conflict subproblems (Gerlach et al. 2025). A robotics-oriented time-indexed QUBO can combine logical preprocessing, adaptive penalties, and time-window decomposition for multi-robot grid planning (González Villasmil 2026). These works reinforce two lessons: path-planning encodings can grow rapidly with space-time discretization, and solver outputs must be interpreted relative to hard feasibility and strong classical references.

Among the works reviewed above, we did not find a study combining all ingredients evaluated here: a single-UAV time-indexed HUBO with a static line-of-sight cost, a genuinely cubic sustained near-obstacle temporal term, explicit HUBO-to-QUBO reduction accounting, decoded feasibility/goal/full-success reporting, and an exact CP-SAT/MILP reference evaluated on the same native objective. The contribution is therefore not a new claim of quantum advantage, but a controlled benchmarking framework designed to expose when apparently favorable binary-optimization energies do or do not correspond to executable UAV paths.

Discrete Planning Model

Let $G = (V, E)$ be an $N \times N$ grid graph. In the experiments $N = 8$, $|V| = 64$, and time layers are indexed $t = 0, \dots, L - 1$ with $L = 20$. Each layer is one discrete trajectory frame; a larger L gives the planner more room for detours or hovering but increases the number of variables linearly. Let $O \subset V$ be the obstacle set, $s \in V$ the start cell, and $g \in V$ the target cell. The permitted next-cell set is

$$\mathcal{N}(u) = \{u\} \cup \{v \in V : (u, v) \in E\}, \quad (1)$$

where the self-loop represents hovering.

The binary state variable is

$$x_{t,v} \in \{0, 1\}, \quad x_{t,v} = 1 \iff v_t = v. \quad (2)$$

Thus $x_{t,v}$ is one exactly when the UAV occupies cell v at layer t . A valid binary solution decodes to a trajectory $\pi = (v_0, \dots, v_{L-1})$ only if exactly one state variable is active at every layer.

Let $\text{LOS}(u, g)$ denote the rasterized Bresenham line cells strictly between u and g . The per-cell visibility/occlusion-depth surrogate is

$$C_{\text{occ}}(u) = |\text{LOS}(u, g) \cap O|. \quad (3)$$

A zero value indicates clear line of sight in this grid model. Let $B \subset V \setminus O$ be the set of non-obstacle cells within Chebyshev distance one of an obstacle. Cells in B are legal but risky.

HUBO Formulation

The total polynomial objective is

$$H(x) = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}}. \quad (4)$$

Quantity	Value
Grid size	8×8
Horizon L	20
Original binary variables	1,280
Total polynomial terms	118,344
Linear terms	1,280
Quadratic terms	112,672
Cubic terms	4,392
Obstacles	6
Buffer cells	24

Table 1: Updated main HUBO instance.

The structural terms are

$$H_{\text{uniq}} = \lambda_u \sum_{t=0}^{L-1} \left(\sum_{v \in V} x_{t,v} - 1 \right)^2, \quad (5)$$

$$H_{\text{start}} = \lambda_s (1 - x_{0,s}), \quad (6)$$

$$H_{\text{move}} = \lambda_m \sum_{t=0}^{L-2} \sum_{u \in V} \sum_{v \notin \mathcal{N}(u)} x_{t,u} x_{t+1,v}, \quad (7)$$

$$H_{\text{obs}} = \lambda_o \sum_{t=0}^{L-1} \sum_{v \in O} x_{t,v}. \quad (8)$$

The static visibility cost is linear:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t=0}^{L-1} \sum_{u \in V} C_{\text{occ}}(u) x_{t,u}. \quad (9)$$

The sole cubic term is the sustained-proximity penalty:

$$H_{\text{prox}} = \lambda_p \eta \sum_{t=1}^{L-2} \sum_{u \in B} \sum_{v \in B \cap \mathcal{N}(u)} \sum_{w \in B \cap \mathcal{N}(v)} x_{t-1,u} x_{t,v} x_{t+1,w}. \quad (10)$$

Here $\eta = 3$. This term penalizes remaining in connected buffer cells for three consecutive layers. Thus the model is genuinely HUBO, but the higher-order degree comes from temporal buffer-risk modeling rather than static visibility.

The goal terms are

$$H_{\text{goal}} = \lambda_g \sum_{t=0}^{L-1} \sum_{v \in V} \frac{t+1}{L} \|p_v - p_g\|_2 x_{t,v}, \quad (11)$$

$$H_{\text{term}} = \lambda_{\text{term}} \sum_{v \neq g} x_{L-1,v}. \quad (12)$$

The updated instance uses $\lambda_u = \lambda_s = \lambda_m = \lambda_o = 1000$, $\lambda_{\text{occ}} = 10$, $\lambda_p = 5$, $\lambda_g = 8$, $\lambda_{\text{term}} = 50$, and $\eta = 3$.

Feasibility-Aware Evaluation

A solver output is not scored only by total HUBO energy. Each binary sample is decoded and checked directly. We report structural feasibility, meaning one-hot occupancy, correct start, legal transitions, and no obstacle occupancy; goal

arrival, meaning $v_{L-1} = g$; and full-task success, meaning both conditions hold for the same decoded sample. Soft cost and native HUBO energy are then interpreted on valid decoded paths.

This distinction is essential for QUBO/BQM and QAOA-style workflows. A sample can have a low reduced-model energy while failing one-hot, motion, obstacle, or target-arrival checks. Such a sample is not a UAV plan. The benchmark therefore reports feasibility rates and representative decoded trajectories, not energy alone.

Solvers and Implementation

A* and RRT Baselines

A* searches directly on the grid with obstacle avoidance and a heuristic path cost. RRT-best is a randomized grid planner used as a sampling-style reference. These methods return explicit paths and are checked with the same feasibility evaluator. The RRT-best runtime includes 40 fixed-seed trials and best-cost selection.

QUAV-Style QAOA Baseline

QUAV maps UAV path segments to qubits and applies QAOA-style optimization with obstacle-buffer costs. Since no official public code was available in this workflow, we implemented a QUAV-style candidate-path selector following the published algorithmic description. It generates collision-free candidate paths, assigns a geometric cost using length, buffer steps, and turns, and uses a small QAOA-style one-hot selector over the candidates. This baseline is a same-platform reference only, not an exact reproduction of the authors' implementation.

Trajectory-Level SA and `neal`

Trajectory-level SA searches over cell sequences and evaluates the native HUBO energy directly, including the cubic term. Its proposal operator preserves one-hot occupancy, start, legal motion, and obstacle avoidance by construction. In contrast, `neal` is a classical BQM simulated-annealing sampler (D-Wave Systems Inc. 2026b). Therefore the HUBO is first reduced to a QUBO/BQM using `dimod` (D-Wave Systems Inc. 2026a); cubic monomials require auxiliary-variable quadratization following the standard reduction principle of Rosenberg (1975) before sampling.

Exact MILP and CP-SAT Baselines

We added exact optimization baselines in both CP-SAT and MILP form. Both encodings enforce the path constraints directly: exactly one cell per layer, fixed start, final target arrival, obstacle cells forbidden, and legal transitions. Auxiliary binary variables linearize each cubic proximity monomial $y_{tuvw} = x_{t-1,u} x_{t,v} x_{t+1,w}$. The objective minimizes

$$H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}} \quad (13)$$

over the hard-feasible set. Because all hard constraints are enforced, every feasible solution has the same hard contribution $-\lambda_u L - \lambda_s = -21,000$ under the expanded HUBO implementation. The native HUBO energy is therefore this

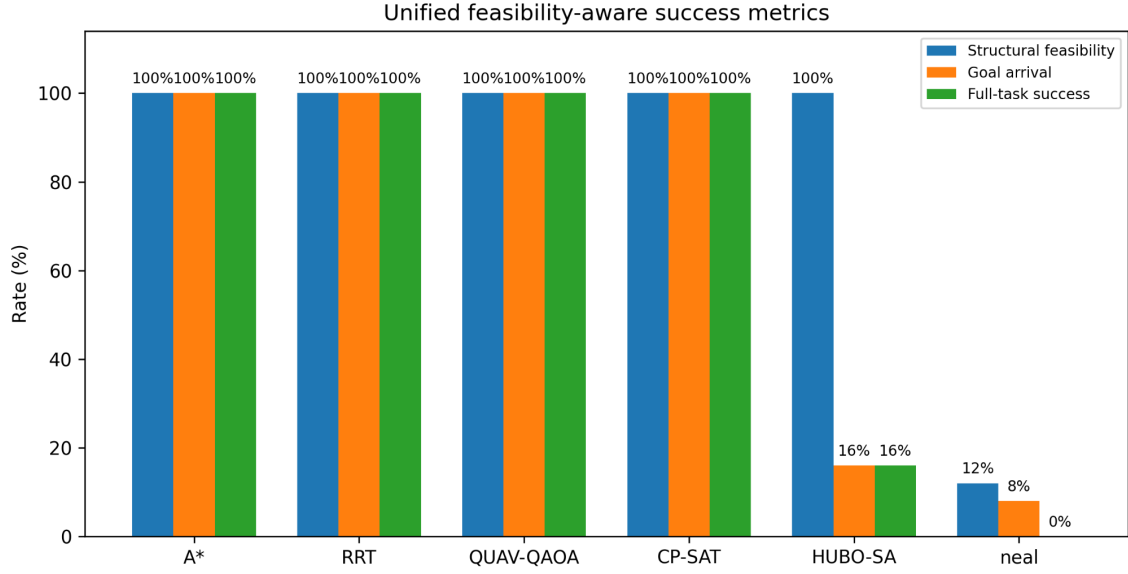


Figure 1: Unified feasibility-aware success metrics. Exact CP-SAT certifies a full-success optimum. The `neal` sampler has nonzero structural-feasibility and goal-arrival rates separately, but zero full-task success.

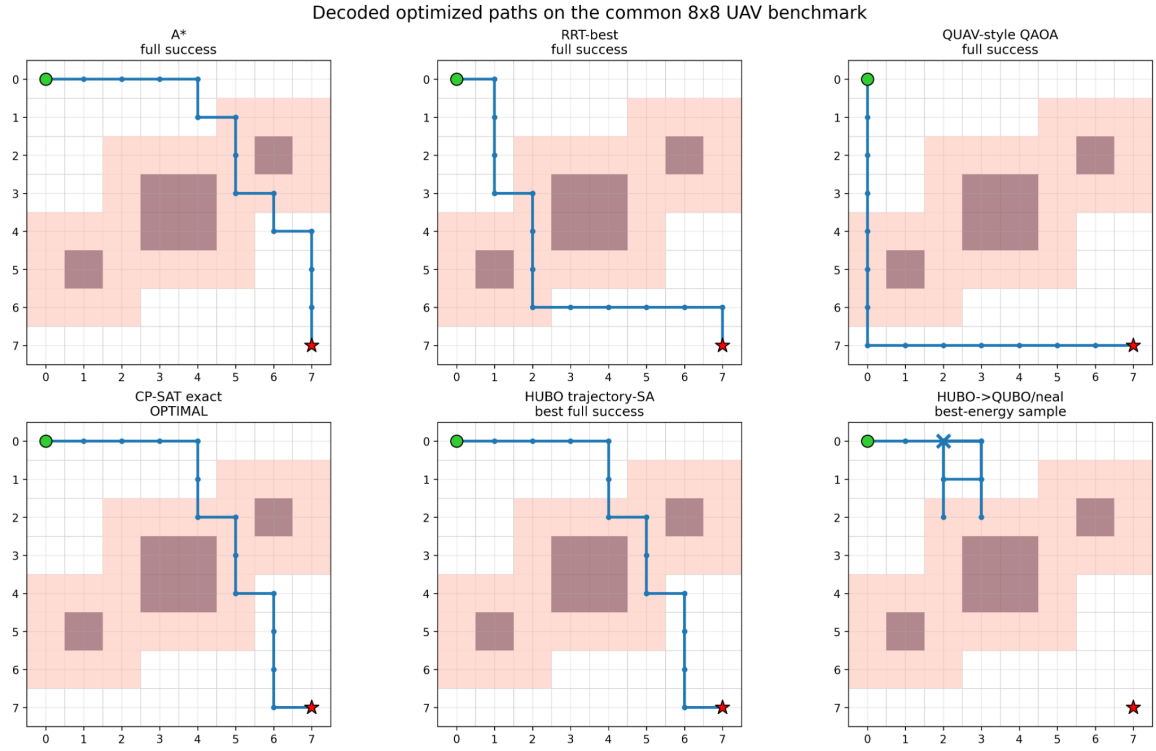


Figure 2: Decoded optimized paths on the common 8×8 , $L = 20$ benchmark. Dark cells are obstacles, light cells are obstacle-buffer cells, the green circle is the start, and the red star is the target. The CP-SAT path is globally optimal for the hard-feasible native-HUBO soft objective. The `neal` path is the best-energy decoded QUBO sample; the cross marks its terminal cell and illustrates a quadratization/sampling failure mode because the path does not reach the target.

hard constant plus the optimized soft cost. The CP-SAT model uses OR-Tools (Perron and Furnon 2026), and the

Method	Results
A*	Feas. Y/Y/Y; path 14/7/4/85.0%; $H = -20,742.26$; time = 0.028 s
RRT-best	Feas. Y/Y/Y; path 14/5/4/70.0%; $H = -20,700.19$; time = 0.043 s
QUAV-QAOA	Feas. Y/Y/Y; path 14/1/3/80.0%; $H = -20,725.94$; time = 4.11 s
CP-SAT exact	Feas. Y/Y/Y; path 14/6/4/85.0%; $H = -20,752.43$; time = 1.02 s
HUBO-SA	Feas. 100.0/16.0/16.0%; path 14/6/4/85.0%; $H = -20,752.43$; time = 1,207.26 s
QUBO/neal	Feas. 12.0/8.0/0.0%; path 18/12/2/20.0%; $H = -20,060.47$; time = 6.15 s

Table 2: Unified same-platform comparison. Feasibility entries are structural/goal/full; path entries are length/turns/buffer/visibility. Stochastic rates use 50 runs or reads.

MILP model uses a linear mixed-integer encoding. The CP-SAT run returned OPTIMAL, so its solution is used as the exact baseline in the final table.

Results

Table 1 summarizes the updated instance.

Table 2 gives the unified same-platform comparison. For deterministic or single-output methods, the three success columns indicate whether the returned path satisfies the condition. For stochastic solvers, they report empirical rates over 50 runs or reads.

Decoded Optimized Paths

Figure 2 shows the actual decoded paths returned by each method on the same map. The numerical table alone does not reveal how the methods use the obstacle-buffer region or why different objectives prefer different routes. A*, RRT-best, QUAV-style QAOA, CP-SAT, and the best full-success SA trajectory all reach the target without collisions. The `neal` panel shows the best-energy decoded QUBO sample; it is structurally feasible but fails to reach the target, explaining why its full-task success rate is zero.

The exact CP-SAT baseline reports OPTIMAL, with soft cost 247.57 and native HUBO energy $-20,752.43$ in 1.02 s. This matches the best trajectory-SA energy, showing that SA found a globally optimal path for the hard-feasible exact objective on this instance, but SA required 1,207.26 s over 50 runs and reached the target in only 8 of 50 runs. A* is much faster and fully successful, but its native HUBO energy is slightly worse because it optimizes a graph-search surrogate rather than the exact cubic proximity-aware objective.

The QUAV-style QAOA selector produces a valid shortest path with only one turn and the lowest geometric cost among its candidates. However, under the native HUBO objective its energy is $-20,725.94$, worse than the CP-SAT/SA optimum. A QUAV-style geometric objective and the native HUBO objective are therefore not the same, even when both return collision-free paths. The `neal` sampler after quadratization remains the weakest method in this run: it has 6/50

structural feasibility and 4/50 goal arrival, but 0/50 full-task success.

Penalty Sweep

The penalty-weight sweep for `neal` shows that weak structural penalties permit low-energy but infeasible samples. In the updated run, structural feasibility was 0/15 for $\lambda \in \{50, 100, 300, 600\}$, 4/15 at $\lambda = 1000$, and 5/15 at $\lambda = 2000$. The best feasible soft cost improved from 915.4 to 590.6 between $\lambda = 1000$ and $\lambda = 2000$, still worse than A*, SA, and CP-SAT. Figure 1 reports the final unified success rates.

Discussion

The results show why the paper’s contribution is not a simple claim that quantum computing outperforms classical planning. On this instance, exact CP-SAT proves the optimum of the hard-feasible native-HUBO soft objective, A* is very fast, and trajectory-SA can find the exact optimum but with poor target-arrival reliability. The decoded paths in Figure 2 make the solver differences explicit. The QUAV-style QAOA baseline shows that a quantum-assisted geometric selector can produce a valid obstacle-free path on the same platform. However, the native HUBO objective distinguishes visibility, temporal buffer risk, and terminal behavior differently. Same-platform decoded-path evaluation is therefore necessary for fair comparison.

The exact CP-SAT/MILP formulation also clarifies the role of quantum-compatible solvers. If a QUBO sampler is used, it must overcome not only path planning but also one-hot, movement, obstacle, target, and quadratization constraints. The benchmark exposes these failure modes by reporting structural feasibility, goal arrival, and full-task success separately.

The QUAV-style row in Table 2 should be interpreted carefully: it is a candidate-path selector over generated paths, not a native solution of the full $x_{t,v}$ HUBO. This route is practical because feasibility is mostly handled before optimization, but it optimizes over a restricted candidate set. Thus the quantum-compatible components demonstrate formulation and benchmarking pathways, not superiority over A*, CP-SAT, or MILP. Those classical methods remain the correctness anchors. If a quadratic sampler is used, the cubic term must first be reduced to quadratic form with auxiliary variables and additional penalties. This reduction is one of the benchmarked failure modes rather than a completed performance claim.

Limitations and Future Work

The model remains a grid abstraction without quadrotor dynamics, control delays, or camera field-of-view constraints. The target is static, so the visibility term is linear; a moving-target or occlusion-free tracking formulation would require a separate derivation with time-varying target states and hard line-of-sight constraints. No official QUAV implementation is publicly available; therefore, our QUAV-style baseline evaluates the published concept rather than reproducing

the full algorithm. A fair comparison will require a complete implementation of QUAV under the same benchmark conditions. Future work should study moving-target formulations, larger maps, stronger candidate-generation policies, and sensitivity of quadratization penalties.

Conclusion

This paper presents a feasibility-aware HUBO/QUBO benchmark and solver-analysis framework for UAV obstacle avoidance with visibility cost. The higher-order structure comes from sustained temporal proximity to obstacle-buffer cells. The unified comparison with A*, RRT-best, QUAV-style QAOA, trajectory-SA, `neal`, and exact CP-SAT shows that geometric path quality, native HUBO energy, and full trajectory validity are different metrics. The CP-SAT run certifies the best observed SA path as optimal for the hard-feasible native-HUBO soft objective, while `neal` after quadratization has zero full-task success in the tested reads. The main contribution is therefore a rigorous quantum-compatible benchmark and feasibility-aware evaluation protocol, not a quantum-speedup claim.

Code Availability

The code, figures, result logs, and benchmark implementations are available in the project repository: <https://github.com/alimohammadi44/uav-hubo-qiml-2026/tree/main>.

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